

Hypothesis testing

A decision about a population, based on a sample

Mucyo Karemera · Stéphane Guerrier

Hypotheses and decisions

Confidence intervals and p -values

Interpreting a study

Does the drug have an effect?

The JAMA trial compared dexamethasone with placebo.

Dexamethasone

Proportion of patients whose sore throat has resolved

Placebo

Proportion of patients whose sore throat has resolved

The outcome and the time of measurement define what “an effect” means.

Does the drug have an effect?

The JAMA trial compared dexamethasone with placebo.

Dexamethasone

Proportion of patients whose sore throat has resolved

Placebo

Proportion of patients whose sore throat has resolved

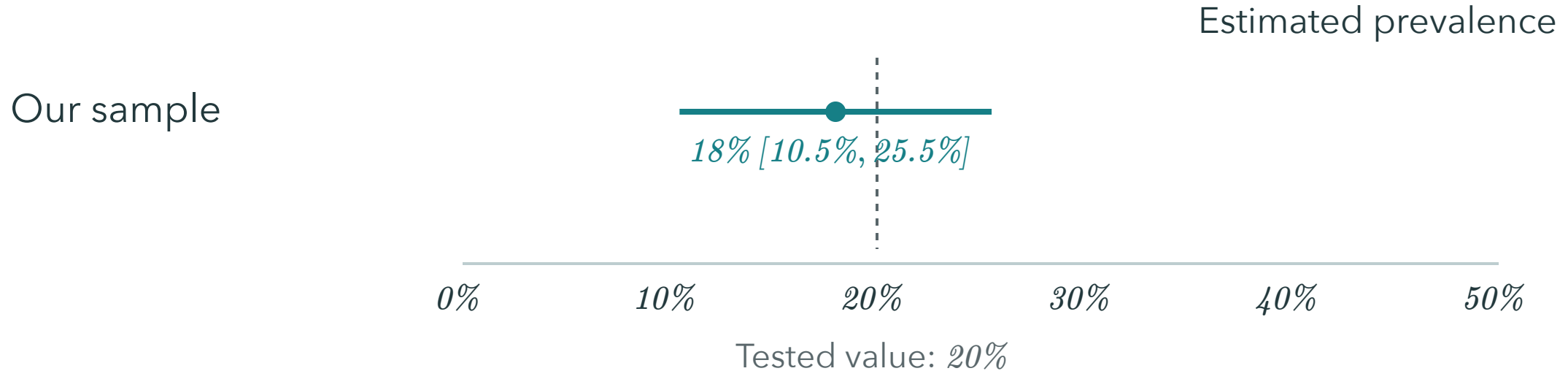
H_0 : no difference between the treatments.

H_1 : a difference between the treatments.

We compare population outcomes using two randomly assigned groups.

Back to our simple prevalence example

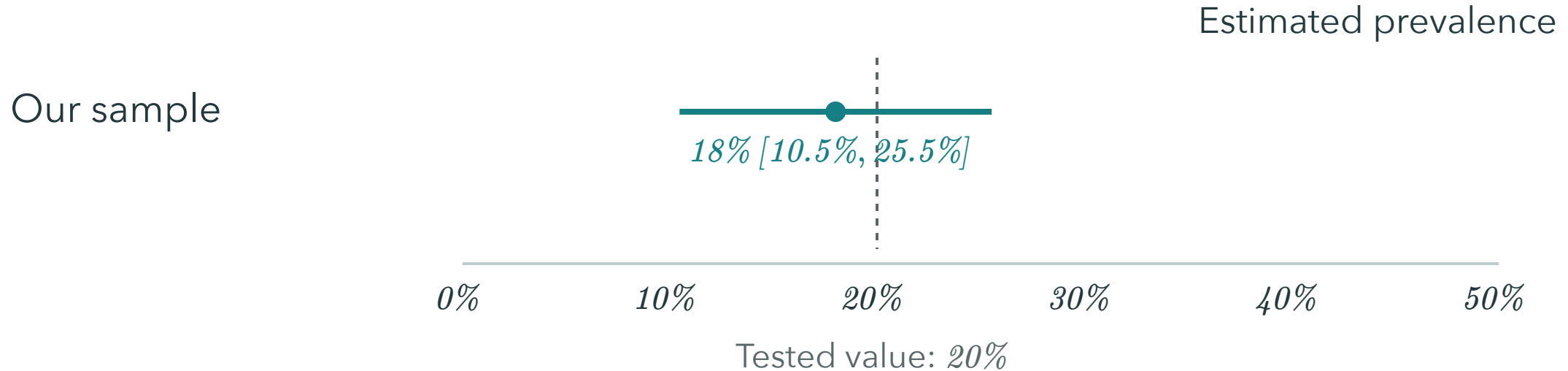
A simpler example: 18 cases among 100 adults.



Estimated prevalence: 18%. An approximate 95% interval: [10.5%, 25.5%].

Back to our simple prevalence example

A simpler example: 18 cases among 100 adults.



How can we turn this comparison into a decision rule?

Hypotheses and significance level

Our question concerns the population prevalence p .

Null hypothesis

$$H_0 : p = 0.20$$

Alternative hypothesis

$$H_1 : p \neq 0.20$$

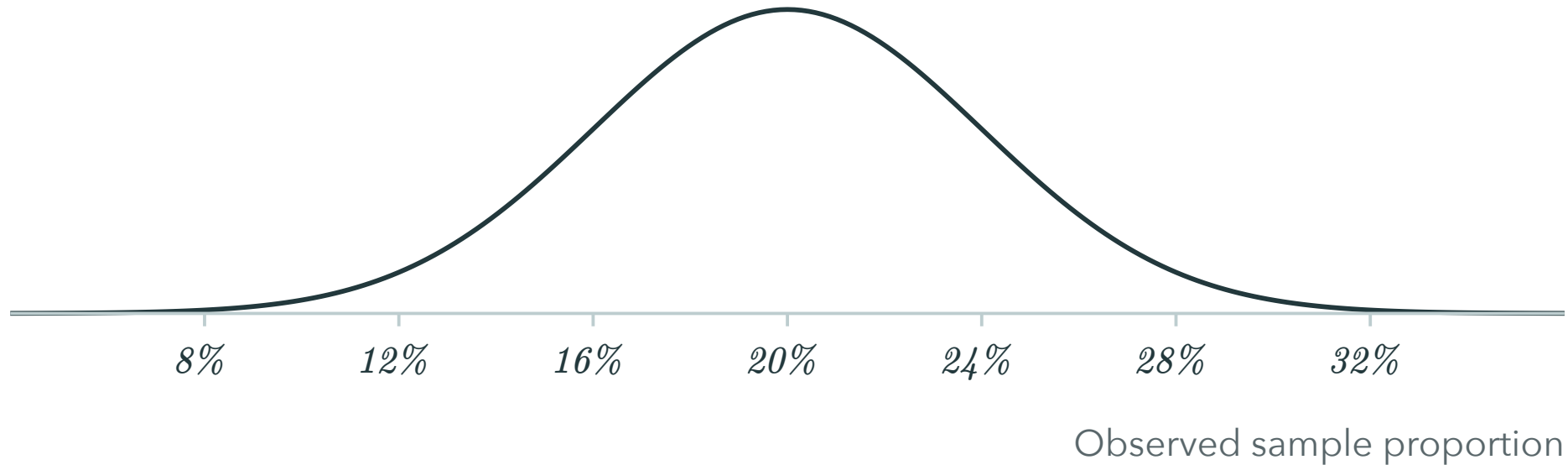
These hypotheses exclude each other.

Choose the significance level in advance: $\alpha = 0.05$.

The significance level sets how extreme a result must be to reject H_0 .

Is our observation unusually far from *20%*?

Suppose H_0 is true. Take samples of *100* adults.

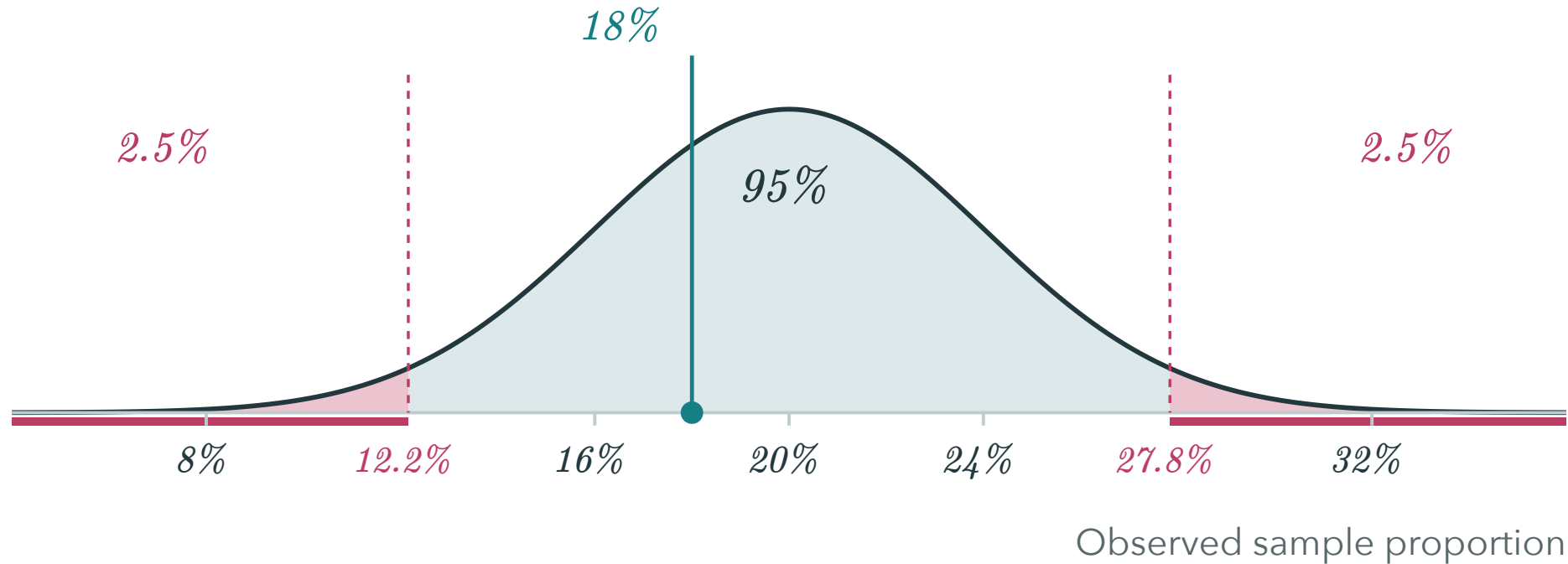


Sample proportions vary around the true prevalence of *20%*.

Normal approximation to the sampling distribution under H_0 .

Is our observation unusually far from 20%?

Suppose H_0 is true. Take samples of 100 adults.

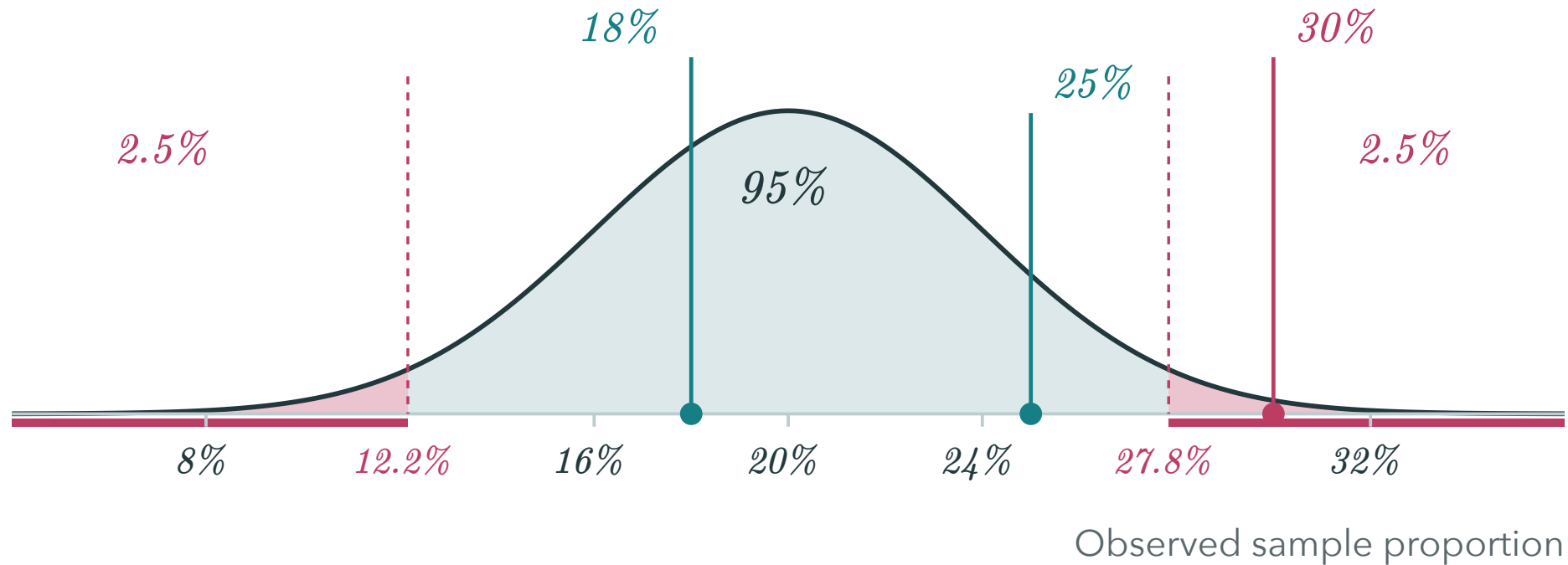


At a 5% significance level, the two extreme regions contain 2.5% each.

Normal approximation to the sampling distribution under H_0 .

Is our observation unusually far from 20%?

Suppose H_0 is true. Take samples of 100 adults.

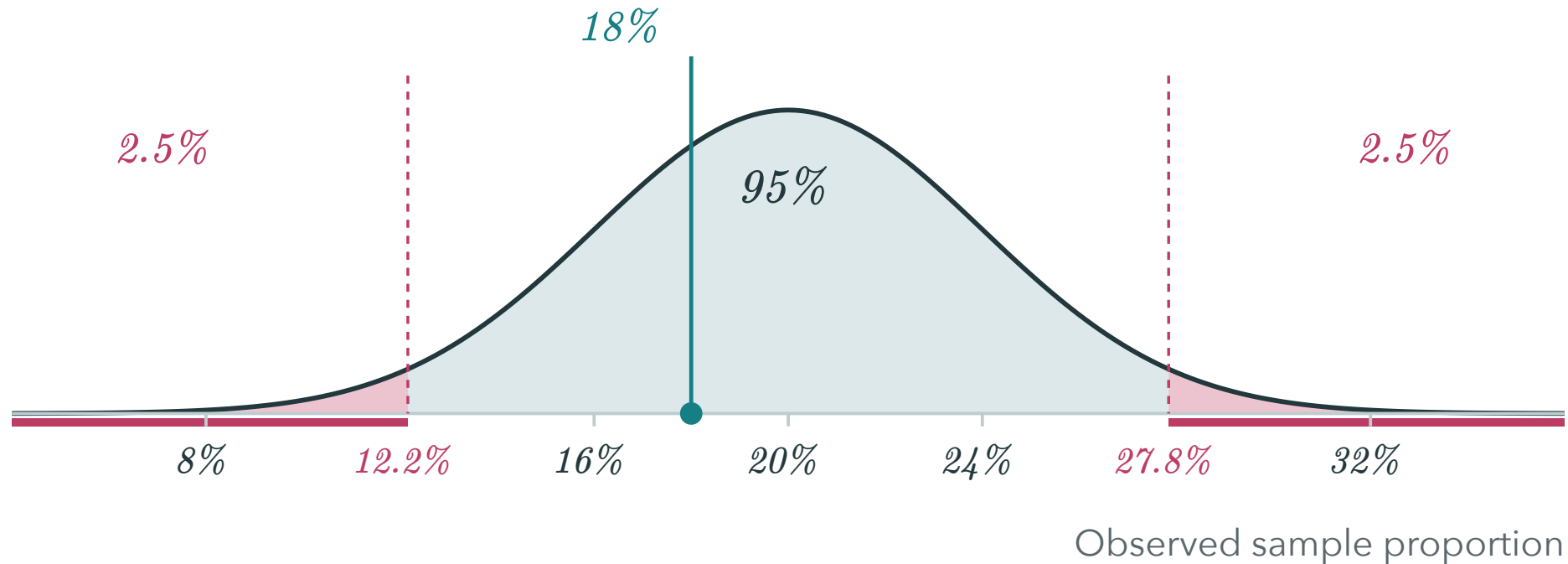


30% falls in an extreme region under H_0 .

Normal approximation to the sampling distribution under H_0 .

One sample, one decision

Same hypothesis and significance level: $H_0 : p = 0.20$, $\alpha = 0.05$.

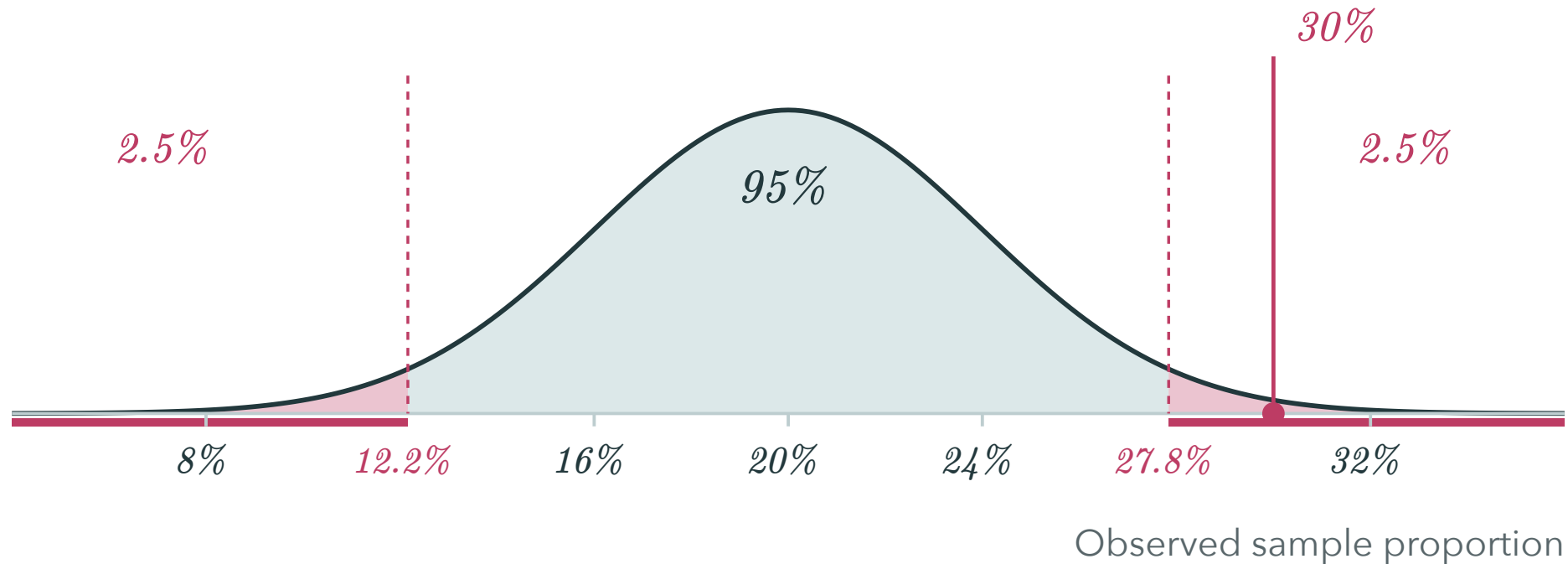


18%: inside the central region. We do not reject H_0 .

Not rejecting H_0 does not establish that the true prevalence is 20%.

One sample, one decision

Same hypothesis and significance level: $H_0 : p = 0.20$, $\alpha = 0.05$.

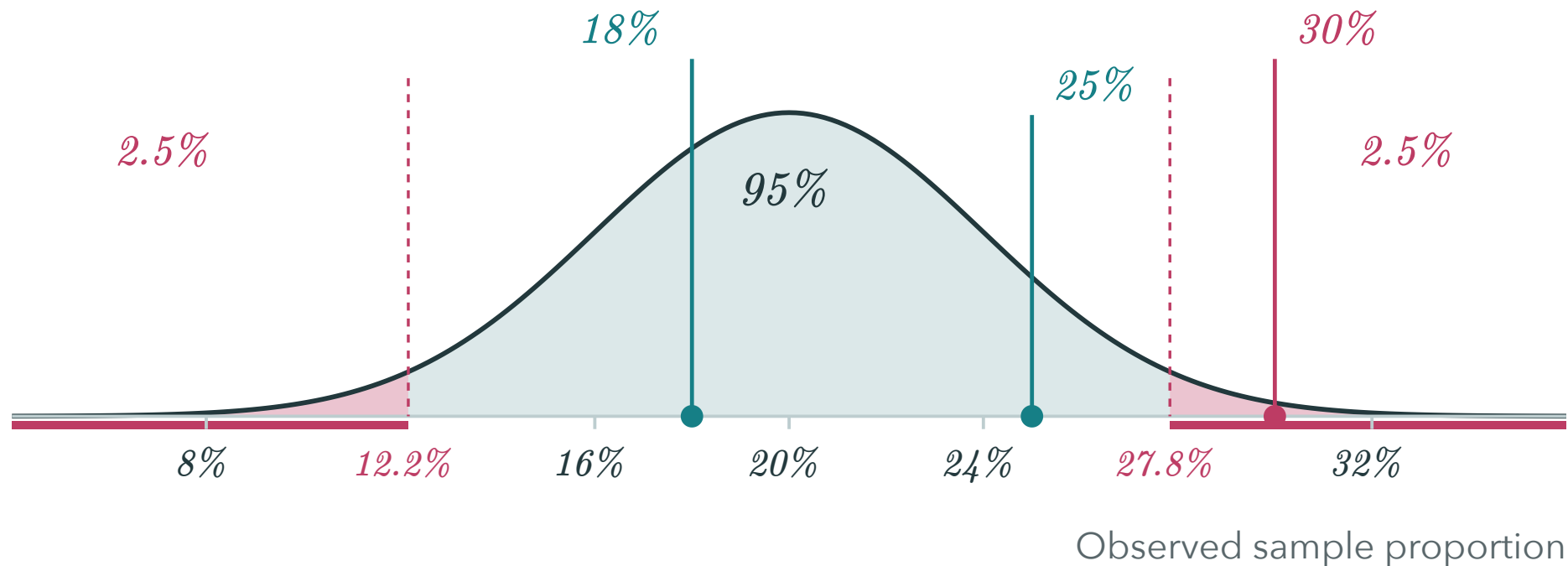


30%: outside the central region. We reject H_0 at 5%.

Not rejecting H_0 does not establish that the true prevalence is 20%.

One sample, one decision

Same hypothesis and significance level: $H_0 : p = 0.20$, $\alpha = 0.05$.

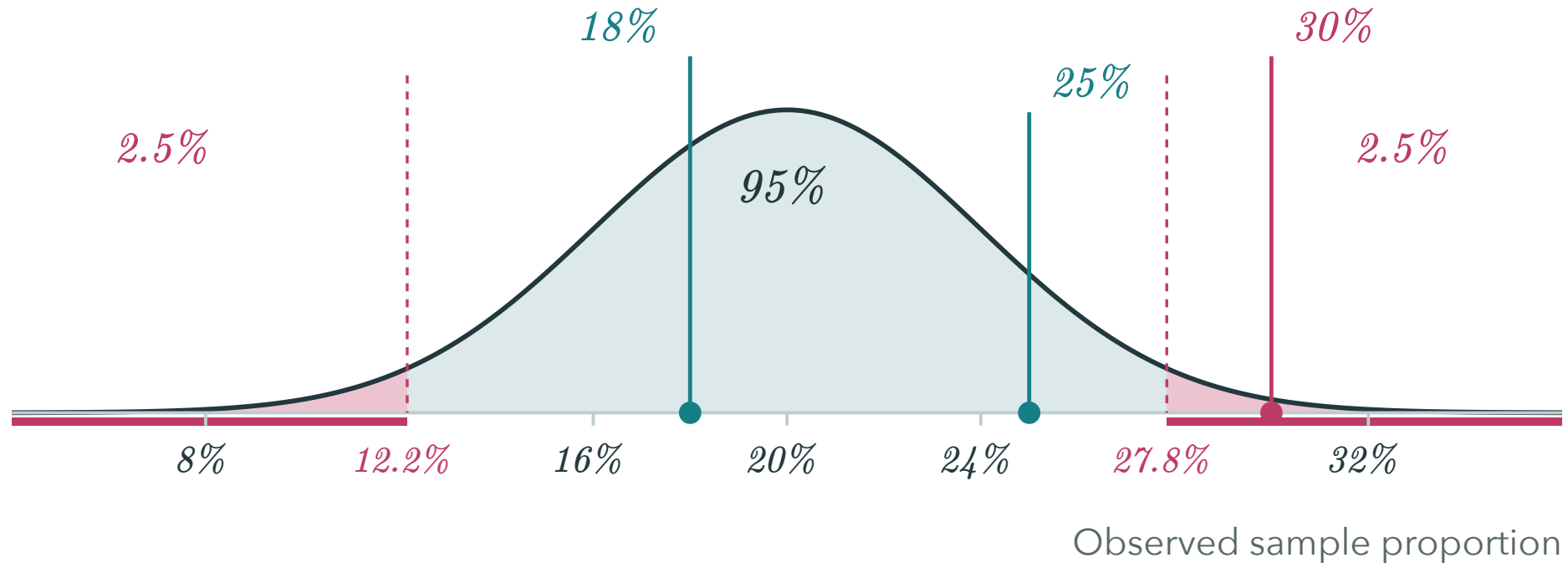


Inside: do not reject H_0 . Outside: reject H_0 at the chosen level.

Not rejecting H_0 does not establish that the true prevalence is 20%.

The significance level changes the rule

Same observations. Significance level: $\alpha = 0.05$.

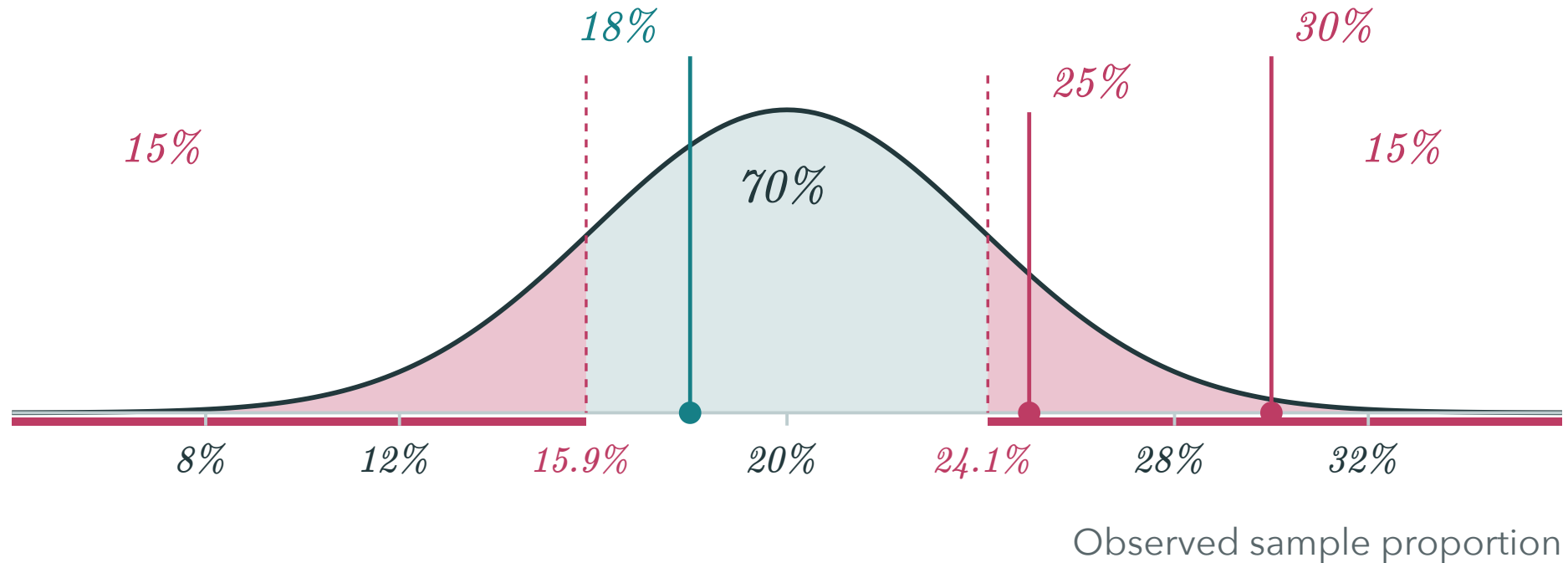


At 5%, only 30% leads to rejection.

Compare rules chosen in advance. The 30% level is deliberately permissive for illustration.

The significance level changes the rule

Same observations. Significance level: $\alpha = 0.30$.

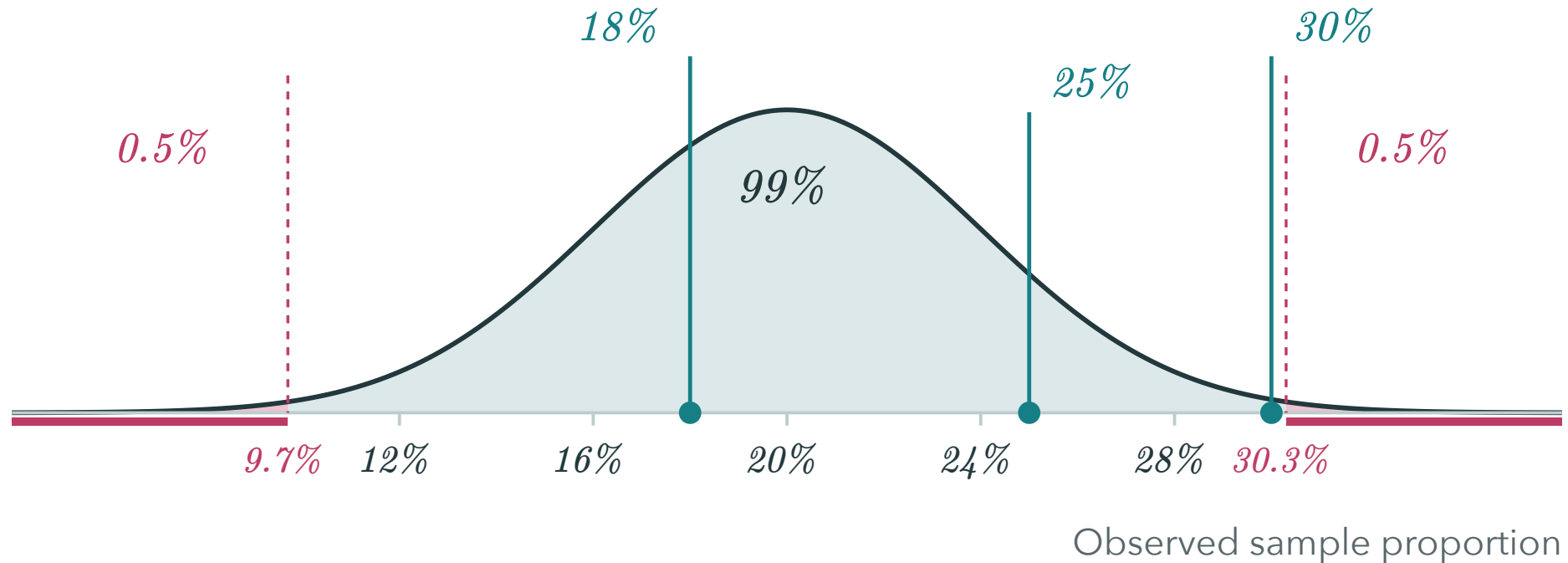


At 30%, both 25% and 30% lead to rejection.

Compare rules chosen in advance. The 30% level is deliberately permissive for illustration.

The significance level changes the rule

Same observations. Significance level: $\alpha = 0.01$.



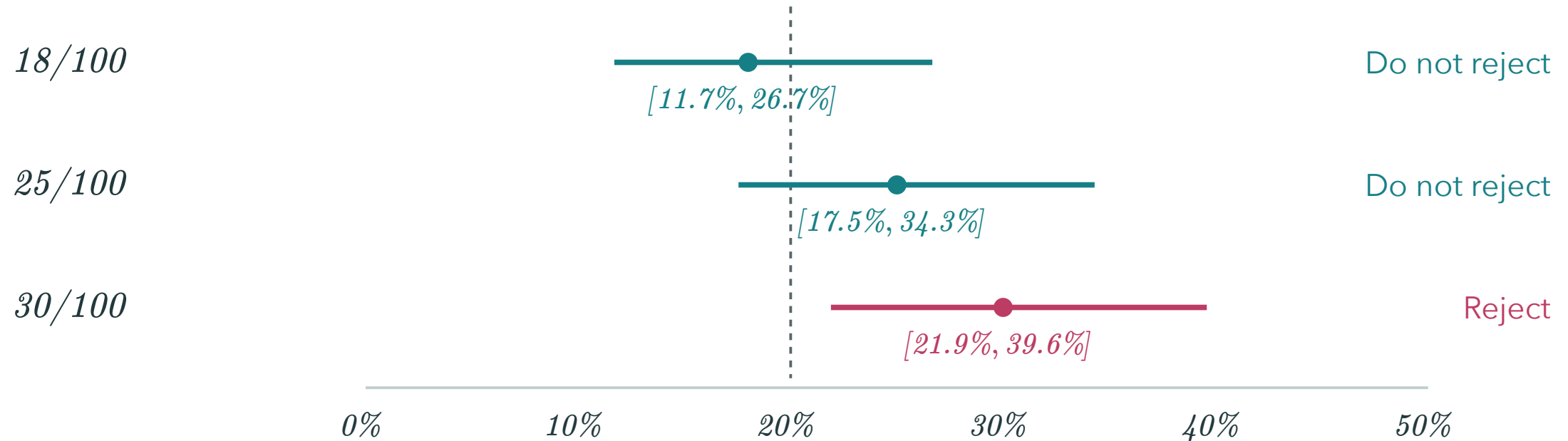
At 1%, none of these three observations leads to rejection.

Compare rules chosen in advance. The 30% level is deliberately permissive for illustration.

Confidence intervals give the matching decision

Two-sided test at 5%. Corresponding confidence level: 95%.

95% confidence intervals

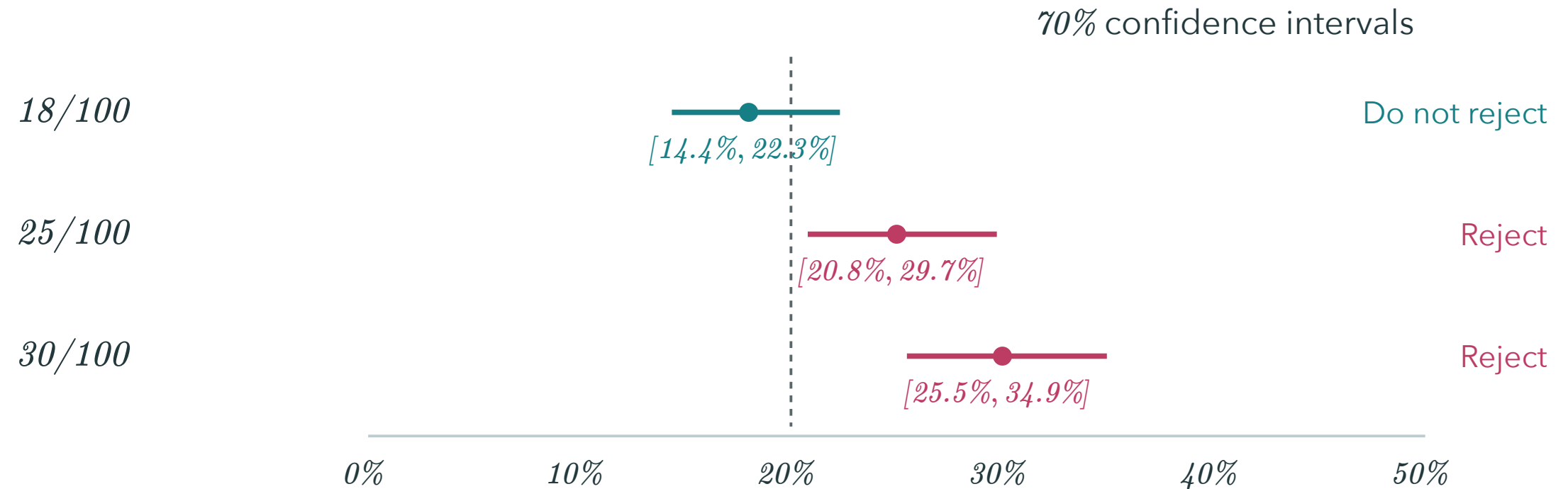


An interval that excludes 20% gives a rejection of H_0 .

These intervals match the test shown here. Greater confidence gives wider intervals.

Confidence intervals give the matching decision

Two-sided test at 30%. Corresponding confidence level: 70%.



An interval that excludes 20% gives a rejection of H_0 .

These intervals match the test shown here. Greater confidence gives wider intervals.

Confidence intervals give the matching decision

Two-sided test at 1% . Corresponding confidence level: 99% .

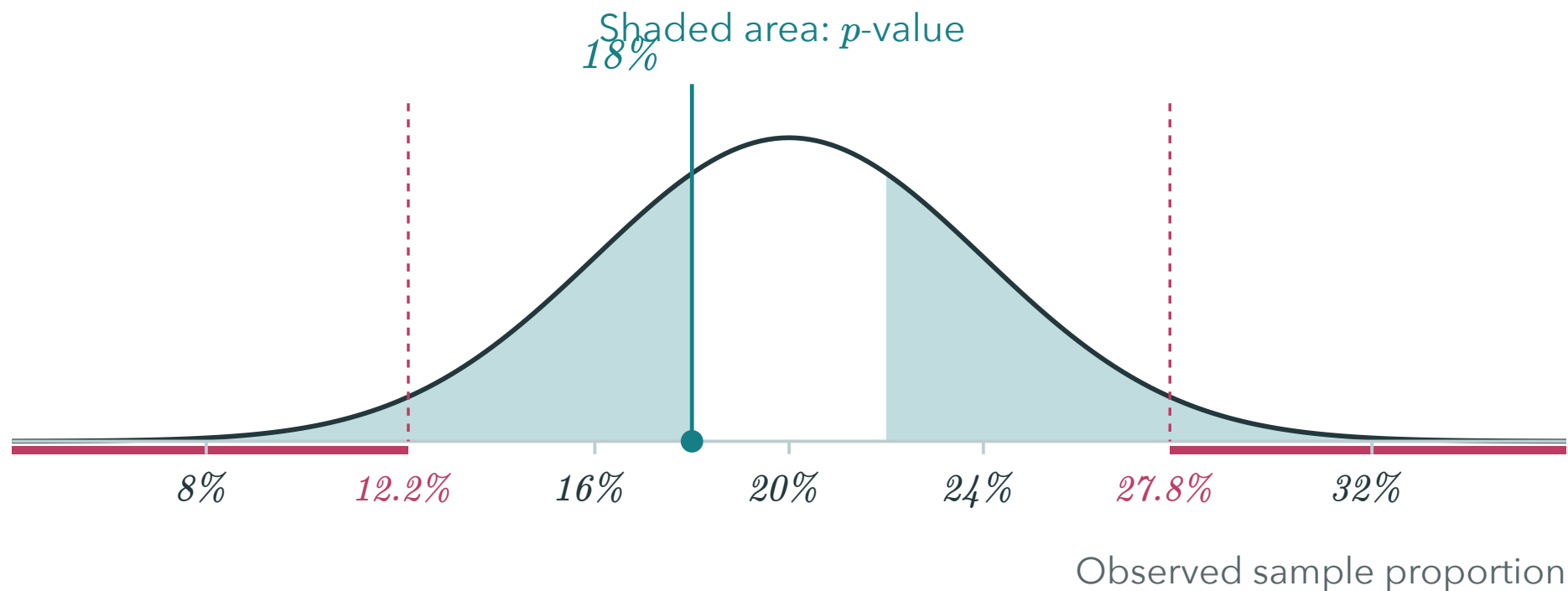


An interval that excludes 20% gives a rejection of H_0 .

These intervals match the test shown here. Greater confidence gives wider intervals.

The p -value measures how extreme the result is

If H_0 were true, how often would a result be at least this far from 20%?

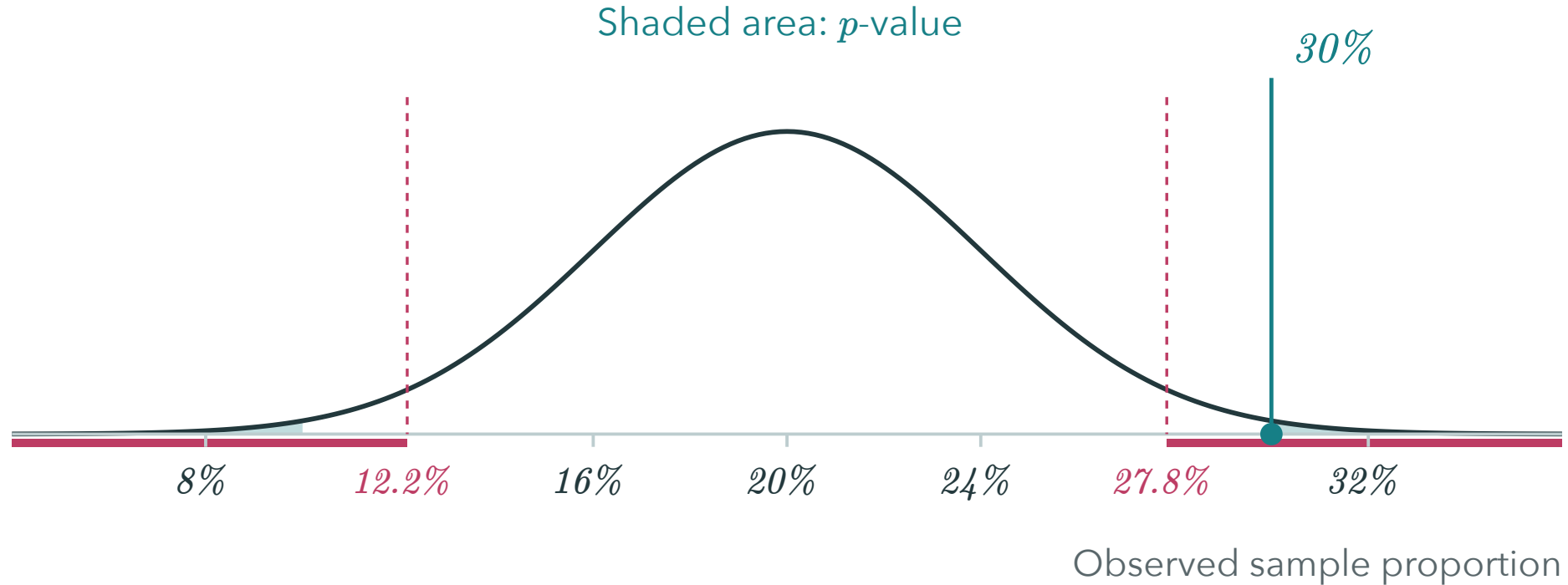


Shaded area: p -value ≈ 0.617 . Do not reject H_0 at 5%.

The p -value follows the observation. The significance level stays fixed.

The p -value measures how extreme the result is

If H_0 were true, how often would a result be at least this far from 20%?



Shaded area: p -value ≈ 0.012 . Reject H_0 at 5%.

The p -value follows the observation. The significance level stays fixed.

The z -score is a way to calculate that area

Express the observed distance in standard errors.

$$z_{\text{obs}} = \frac{\text{estimate} - \text{value in } H_0}{\text{standard error under } H_0}$$

$$18 \text{ cases: } z = -0.50 \qquad 30 \text{ cases: } z = 2.50$$

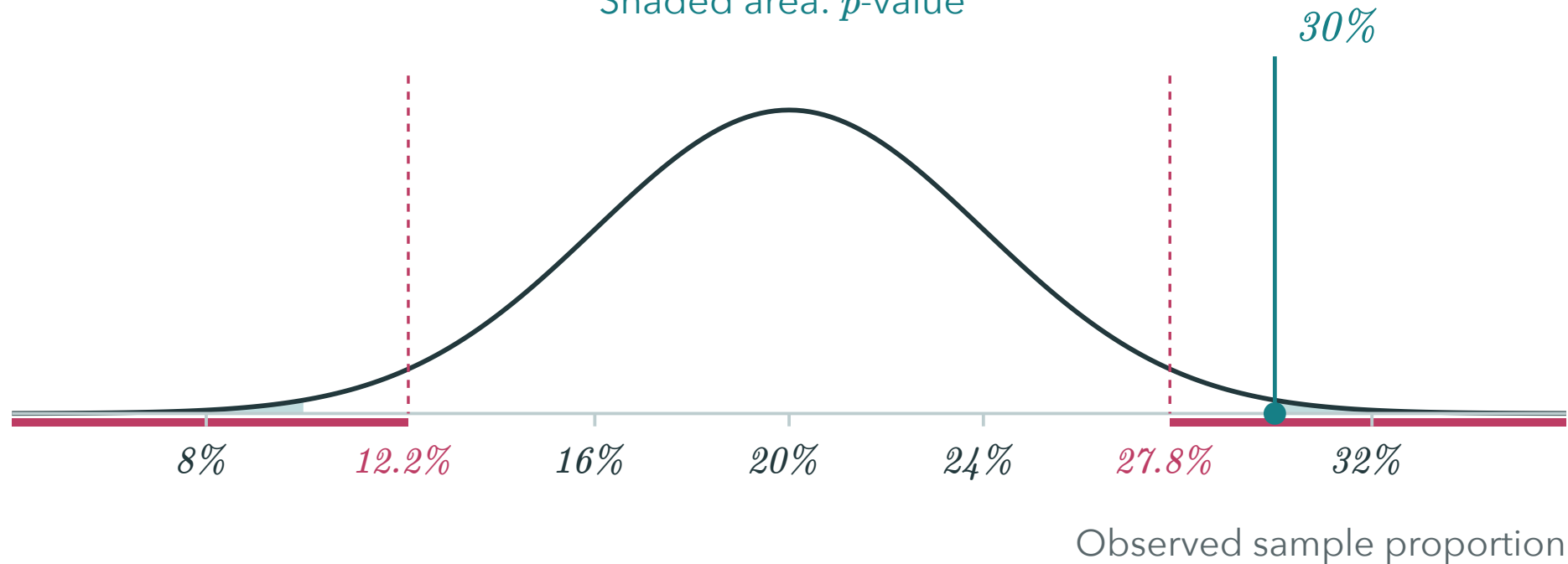
$$\text{two-sided p-value} \approx 2\Pr(Z \geq |z_{\text{obs}}|)$$

Software calculates the area. The graph explains its meaning.

One-sided and two-sided questions

Different from 20%? $H_0 : p = 0.20$, $H_1 : p \neq 0.20$

Shaded area: p-value



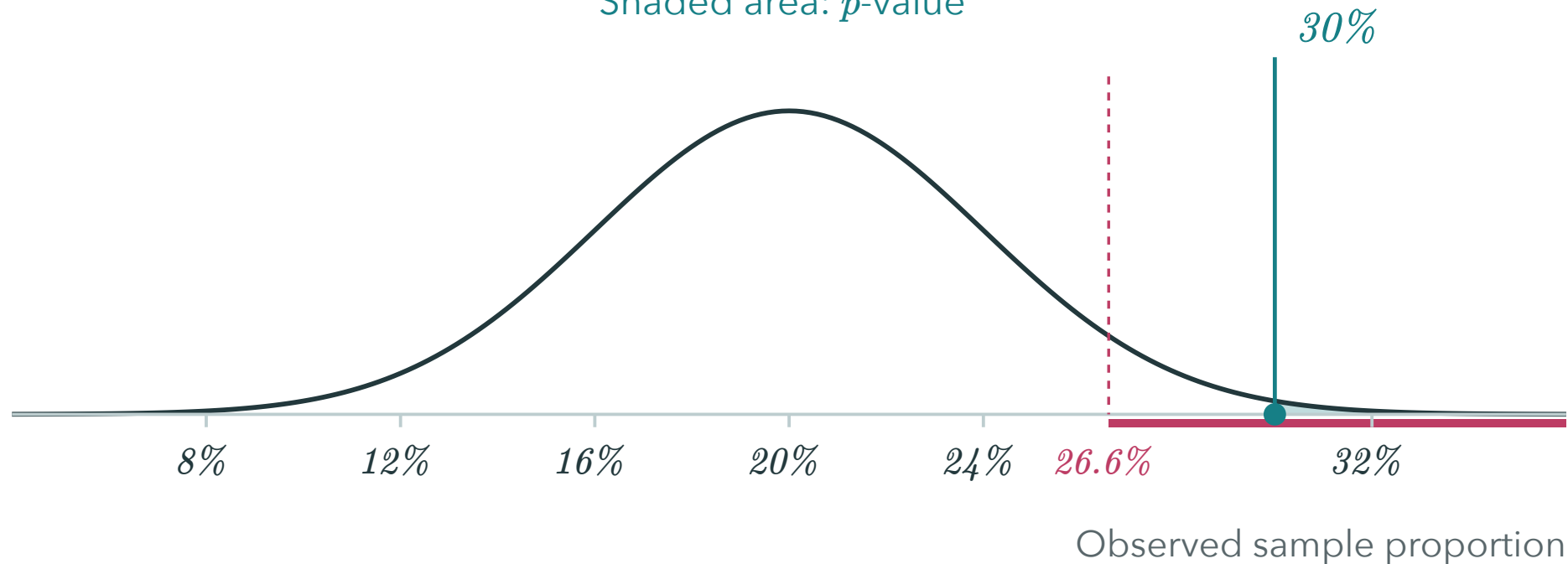
$$\text{p-value} \approx 2\Pr(Z \geq |z_{\text{obs}}|)$$

Choose the direction before seeing the result. The same sample answers different questions.

One-sided and two-sided questions

Higher than 20%? $H_0 : p \leq 0.20$, $H_1 : p > 0.20$

Shaded area: p -value



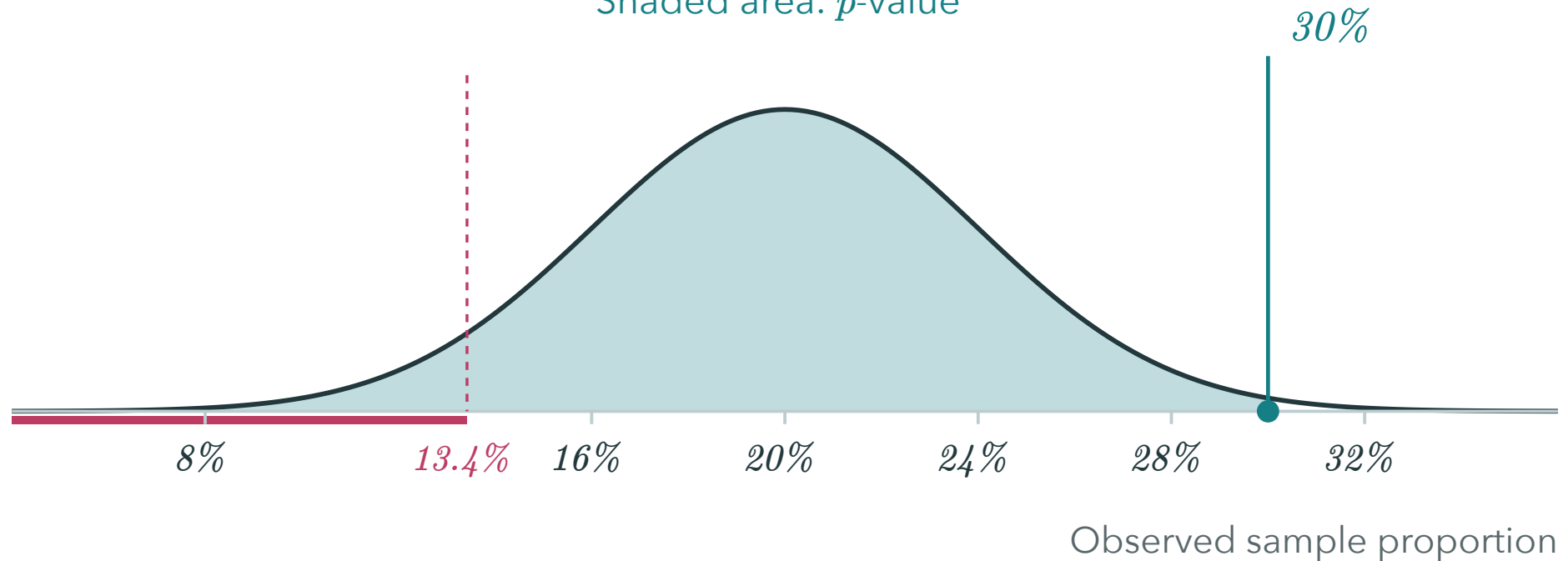
$$p\text{-value} \approx \Pr(Z \geq z_{\text{obs}})$$

Choose the direction before seeing the result. The same sample answers different questions.

One-sided and two-sided questions

Lower than 20%? $H_0 : p \geq 0.20$, $H_1 : p < 0.20$

Shaded area: p -value



$$\text{p-value} \approx \Pr(Z \leq z_{\text{obs}})$$

Choose the direction before seeing the result. The same sample answers different questions.

The same result, different questions

30 cases among 100 adults.

Question	p -value	Decision at 5%
Different from 20%?	0.0124	Reject H_0
Higher than 20%?	0.0062	Reject H_0
Lower than 20%?	0.9938	Do not reject H_0

When the Normal calculation is appropriate

The sample must support the population question.

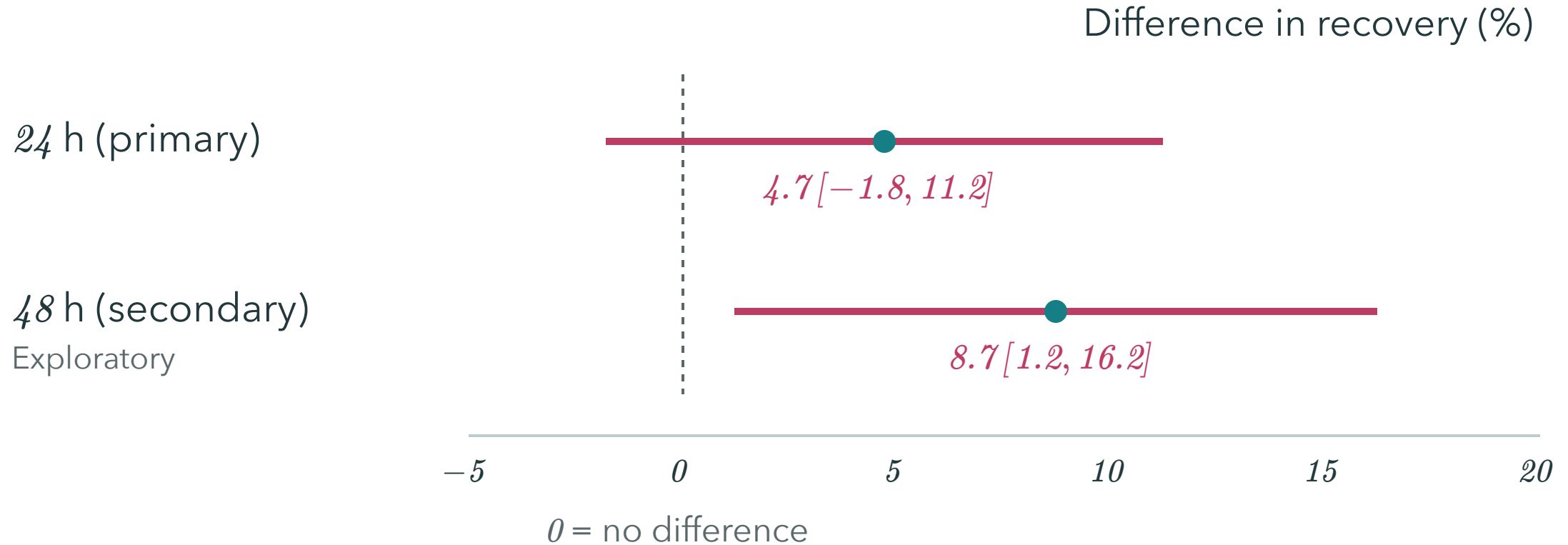
This simple calculation assumes independent observations.

The sample needs enough cases and non-cases for a useful Normal approximation.

Small samples, rare events or dependent data may require another method.

Back to the JAMA confidence intervals

Published differences in symptom resolution: dexamethasone minus placebo.

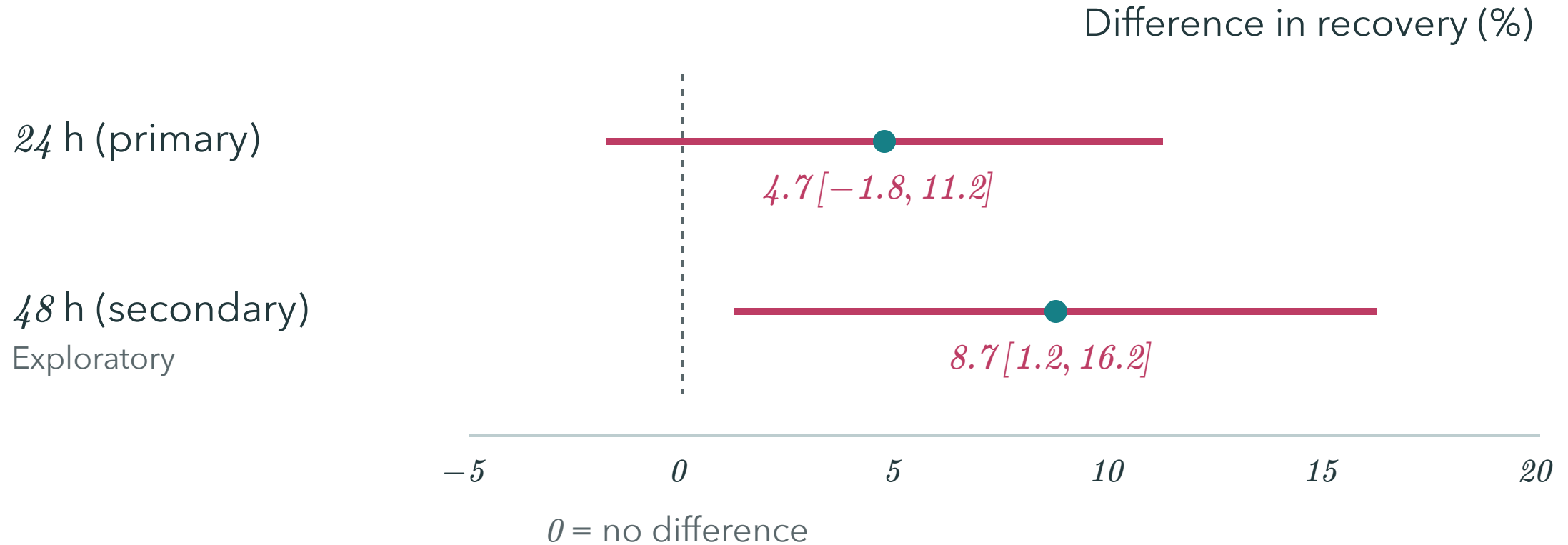


Source: Hayward et al., *JAMA* 2017;317:1535-1543, Figure 2 (redrawn).

No treatment difference corresponds to 0 .

Back to the JAMA confidence intervals

Published differences in symptom resolution: dexamethasone minus placebo.

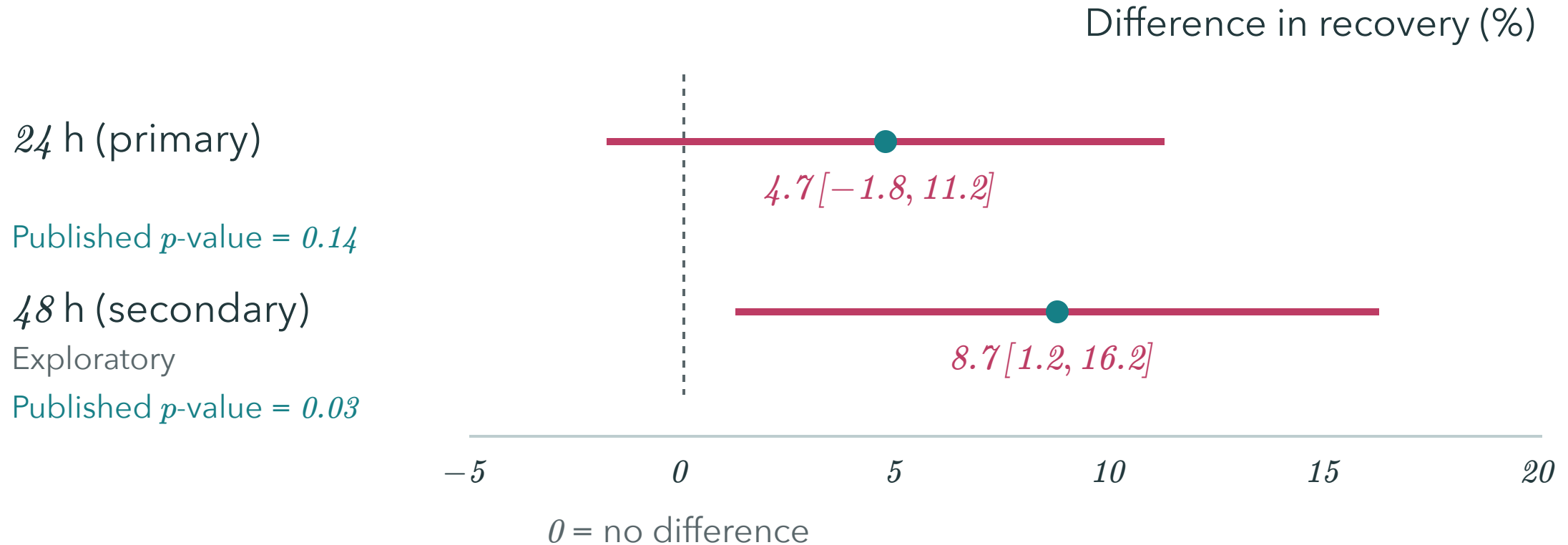


Source: Hayward et al., *JAMA* 2017;317:1535-1543, Figure 2 (redrawn).

At 48 h, the interval excludes 0 and lies on the side favouring dexamethasone.

The published p -values and the conclusion

H_0 : no difference in symptom resolution between treatments.



Source: Hayward et al., *JAMA* 2017;317:1535-1543.

48 h: p -value = 0.03. We reject H_0 at the reported 5% level.

The published p -values and the conclusion

24 h, primary outcome

The study did not demonstrate a statistically significant difference.

48 h, secondary outcome

The study found a difference favouring dexamethasone in an exploratory analysis.

Non-significant does not establish no effect.

Statistically significant does not establish clinical importance.

Source: Hayward et al., *JAMA* 2017;317:1535-1543.

The article reports two-sided tests, without adjustment for multiple comparisons.

References

1. Hayward G, Hay AD, Moore MV, et al. Effect of oral dexamethasone without immediate antibiotics vs placebo on acute sore throat in adults: a randomized clinical trial. *JAMA* 2017;317(15):1535. doi:[10.1001/jama.2017.3417](https://doi.org/10.1001/jama.2017.3417)

Thank you very much for your attention